

BEACON LABS

CYBERSECURITY EDUCATION LABS

Workshop



BOISE STATE UNIVERSITY

COLLEGE OF ENGINEERING

Department of Computer Science

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section*Virtual Machine Setup

Virtual Machine Software Installation

VirtualBox will be the virtual machine software that is used to open the VM images and complete the activity. It is possible to complete the activity with other virtual machine software but it has been untested and so support for other virtual machine software will not be included in this workshop document at this point in time.

Begin by downloading Oracle's VirtualBox software version 7.0.18. The download links for VirtualBox can be found at the below [website](https://www.virtualbox.org/wiki/Download_Old_Builds_7_0). If you already have a newer version of VirtualBox installed, you should be able to follow the same steps.

https://www.virtualbox.org/wiki/Download_Old_Builds_7_0

Windows Setup

Steps:

1. Select the hyperlink titled "Windows hosts" under the "VirtualBox x.x.xx platform packages" as seen in Figure 1 to begin the installation of the latest version of VirtualBox. This activity was created when VirtualBox 7.0.18 was the latest release, however, the steps provided should still apply to future versions of VirtualBox as well.
2. Once the download has completed, run the downloaded installer. The file name should be "VirtualBox-<version_number>-Win.exe".
3. Run through the installer without changing anything. At the end, make sure the checkbox is selected to "Start Oracle VM VirtualBox x.x.xx after installation" as shown in Figure 2 then click the "Finish" button to complete the installation and open VirtualBox.
4. Once VirtualBox has been installed and opened, please continue to the "Downloading and Importing the Images" section.

- **VirtualBox 7.0.18** (released May 07 2024)
 - **Windows hosts**
 - macOS / Intel hosts
 - Solaris hosts
 - Solaris 11 IPS hosts
 - Linux Hosts:
 - Oracle Linux 9 / Red Hat Enterprise Linux 9
 - Oracle Linux 8 / Red Hat Enterprise Linux 8
 - Oracle Linux 7 / Red Hat Enterprise Linux 7 / CentOS 7
 - Ubuntu 24.04
 - Ubuntu 22.04
 - Ubuntu 20.04
 - Ubuntu 18.04 / 18.10 / 19.04
 - Debian 12
 - Debian 11
 - Debian 10
 - openSUSE 15.3 / 15.4 / 15.5
 - Fedora 40 / 41 / 42
 - Fedora 36 / 37 / 38 / 39
 - Fedora 35
 - All distributions
 - Extension Pack
 - Guest Additions
 - Sources
 - MD5 checksums, SHA256 checksums

Figure 1: Download button is highlighted in gray for Windows machines.



Figure 2: Final screen for the Windows installer.

Linux Setup

1. From the VirtualBox download page seen in Figure 1, find your distribution of Linux and follow the instructions Oracle has provided.
2. If your distribution isn't shown on the website, please look into distribution specific instructions for your Linux distro. For example, if you are using Arch Linux, there is a VirtualBox page on the ArchWiki that provides an installation guide.

Downloading and Importing the Images

This activity will be done using two virtual machines that will be provided to you. The image files for this activity's VM can be found in the following [Google Drive folder](https://drive.google.com/drive/folders/19hjc2iZTGFC1OyWLpr8P4DVq1M6V3B9-?usp=sharing), full link shown below.

<https://drive.google.com/drive/folders/19hjc2iZTGFC1OyWLpr8P4DVq1M6V3B9-?usp=sharing>

Steps:

1. Use the above link to open the Google Drive folder containing the two virtual machine image files.
2. Download both image files. The files are 7.79 and 8.61 GB in size so the download may take a while.
3. Once the images have been downloaded, they can be imported into VirtualBox by navigating to File > Import Appliance and selecting the .ova image files that were downloaded. Only one .ova file can be selected and imported at a time. Visual instructions can be seen in Figures 3, 4, 5, 6, and 7.

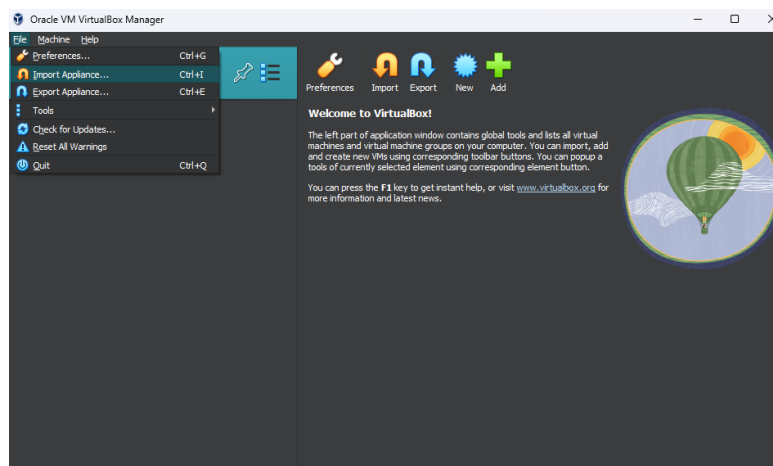


Figure 3: File > Import Appliance

4. Repeat steps 2 and 3 with the other image file.

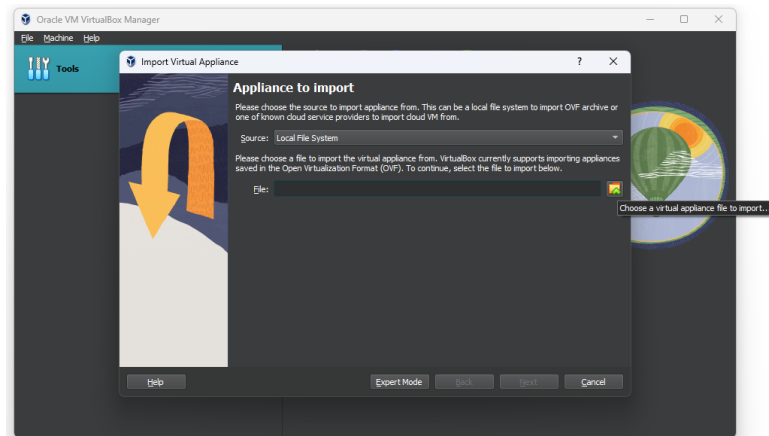


Figure 4: Select the file button to choose one of the downloaded image file.

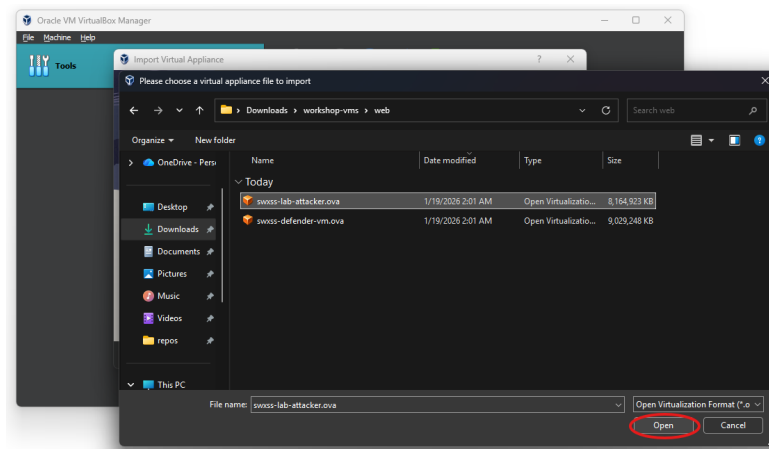


Figure 5: Choose one of the downloaded .ova files to import.

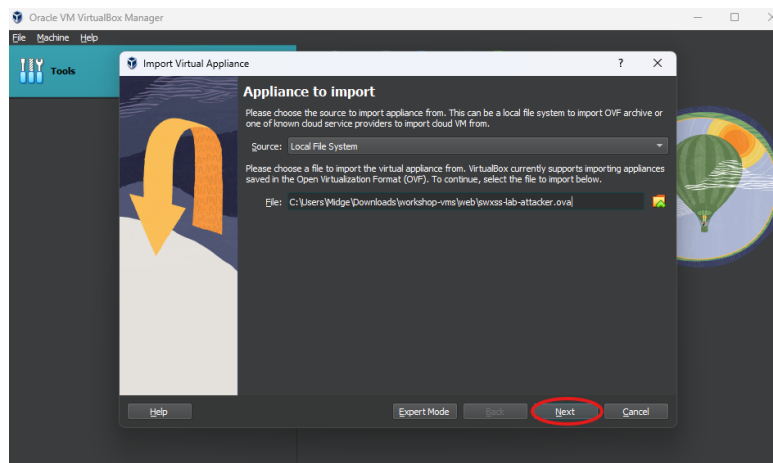


Figure 6: Select next after selecting the image to import.

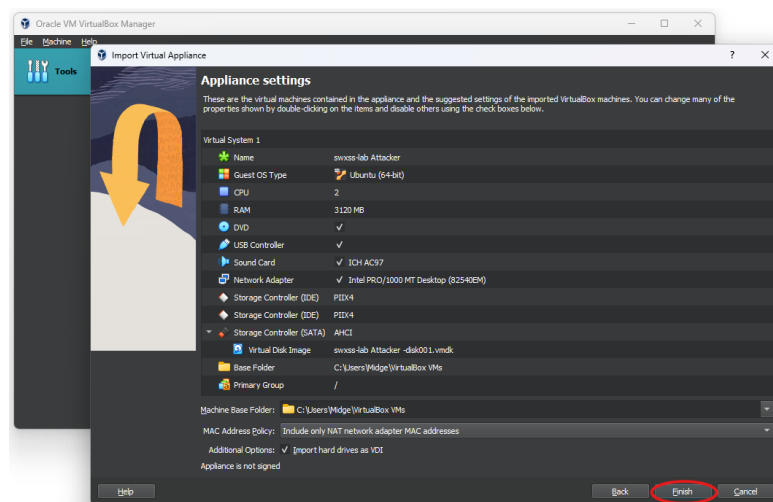


Figure 7: Final confirmation. Your file paths and names may differ slightly.

5. Once both images have been imported to VirtualBox, continue to the final section, “Verifying Import.”

Verifying Import

Steps:

1. Before starting the virtual machines, verify that both images are on the “swxss-network” as seen in Figure 8. If one or both machines are not on this network, skip to Step 6 and then come back here after.

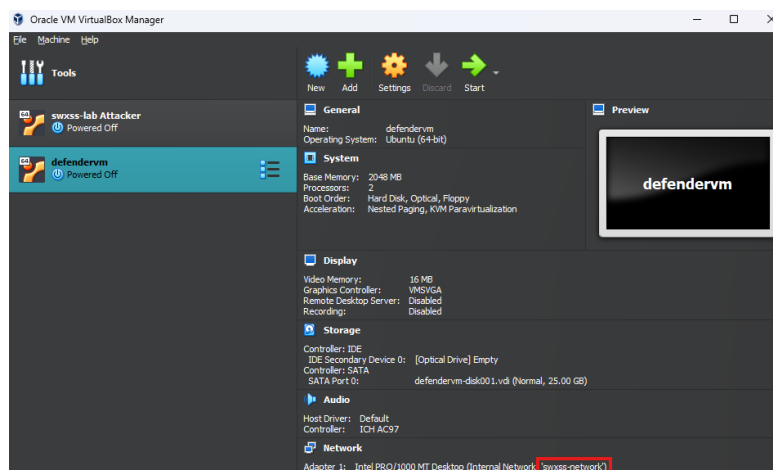


Figure 8: Ensure that the Network is an Internal Network titled “swxss-network.”

2. Start the “defendervm” virtual machine by selecting it and clicking the green “Start” button. The defender vm can be started in the headless state, shown in Figure 9, if desired.
3. Start the “swxss-lab Attacker” virtual machine by selecting it and clicking the green “Start” button.
4. Within the attacker virtual machine, using the username and password “swxss” log in and open up the Firefox browser.
5. Enter the url <https://www.swxss.com/dom-example.php#context=Mary>. If greeted with a the Main Dashboard for Mary as seen in Figure 10 skip to Step 7, otherwise continue on to the next step.

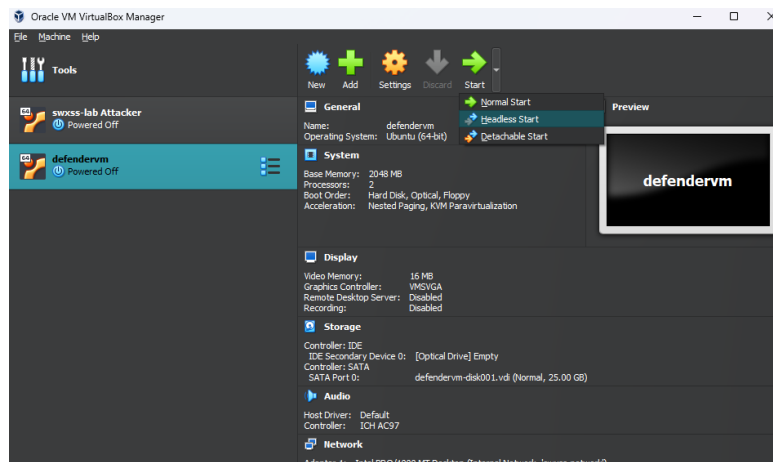


Figure 9: The defender vm can be started in a headless state to have less windows cluttering your desktop.

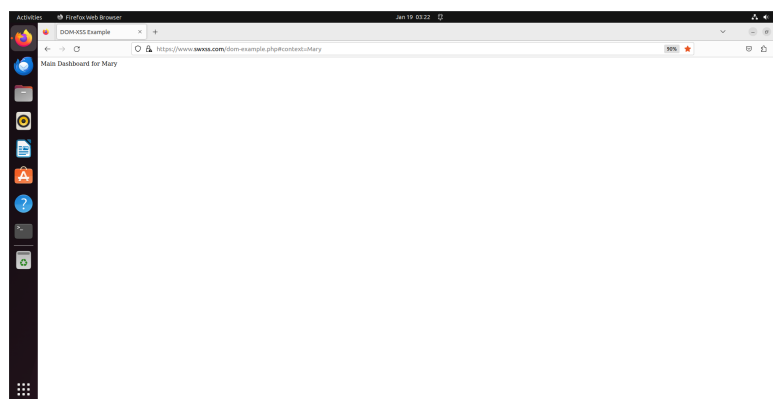


Figure 10: Success!

6. The most likely culprit of the attacker not being able to see the website is the internal network not importing correctly. If one or both of the images are not on the same internal network—as seen in Figure 8—then the website cannot be reached. To put both machines on the same network, first shut down both virtual machines. Once both machines are shut down, select one of them and open the settings (Figure 11). Once the Settings for the machine has been opened, navigate to the Network menu (Figure 12). For each adapter after Adapter 1 in the Network menu, uncheck the “Enable Network Adapter” checkbox (Figure 13). For Adapter 1, make sure the “Enable Network Adapter” checkbox is checked and change the “Attached to:” option to “Internal Network” (Figure 14). Finally, change the name of the internal network to “swxss-network” (Figure 15). Repeat this process for the other machine and then go back to Step 1. If issues persist, please reach out to Dylan Gresham (dylangresham@u.boisestate.edu).

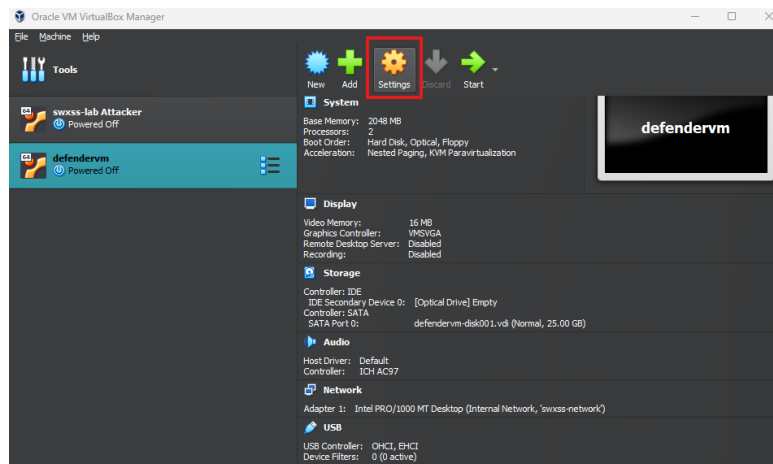


Figure 11: Open settings for a virtual machine.

7. Congratulations, you’ve successfully imported both virtual machine images and are now ready for the Web Security workshop segment of the BEACON Labs Workshop. Thank you for coming prepared! If you haven’t already, please follow the instructions for the Network and Mobile Security material setups. We look forward to seeing you in Boise for the workshop!

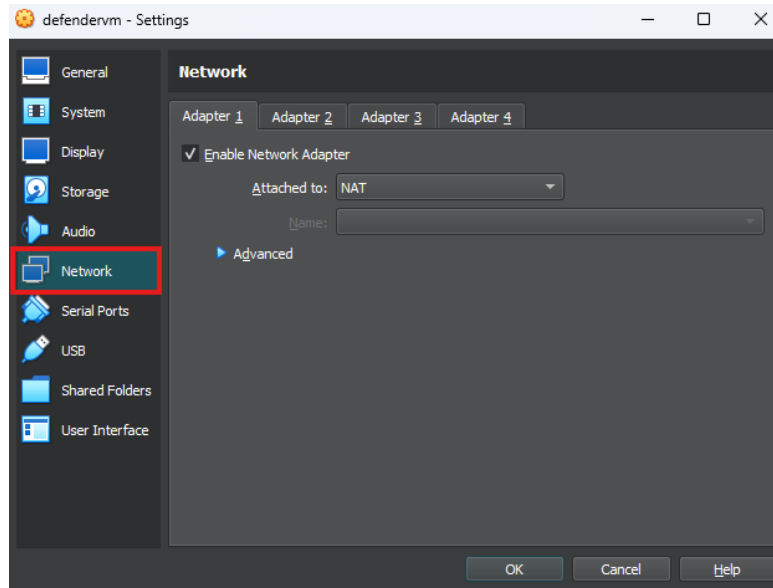


Figure 12: Open the Network settings for a virtual machine.

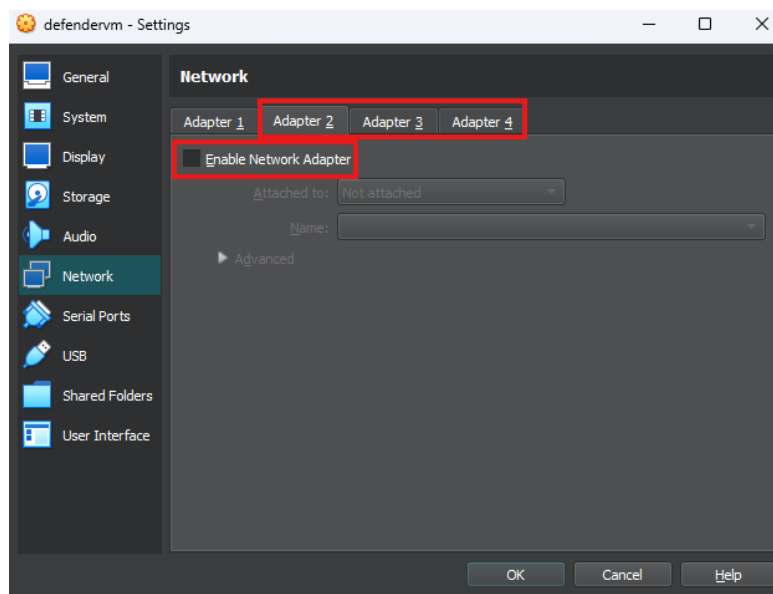


Figure 13: Disable network adapters 2, 3, and 4.

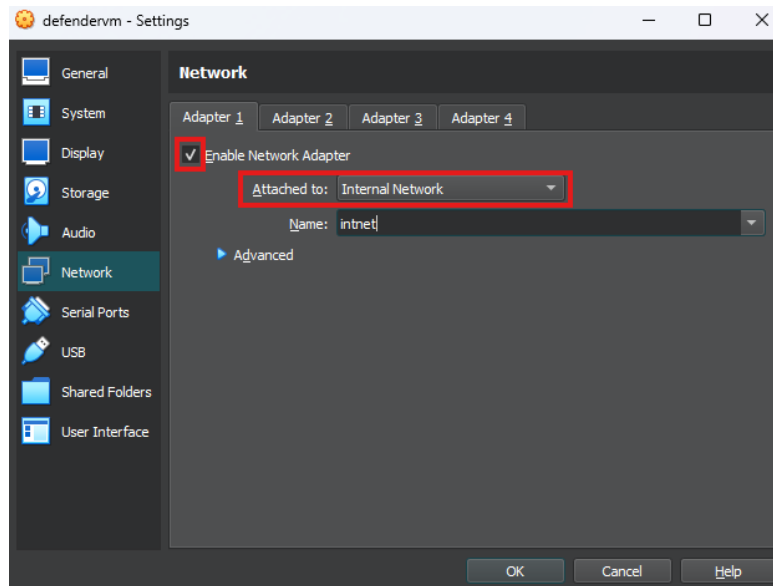


Figure 14: Change Adapter 1 to attach to an Internal Network (name of network may vary).

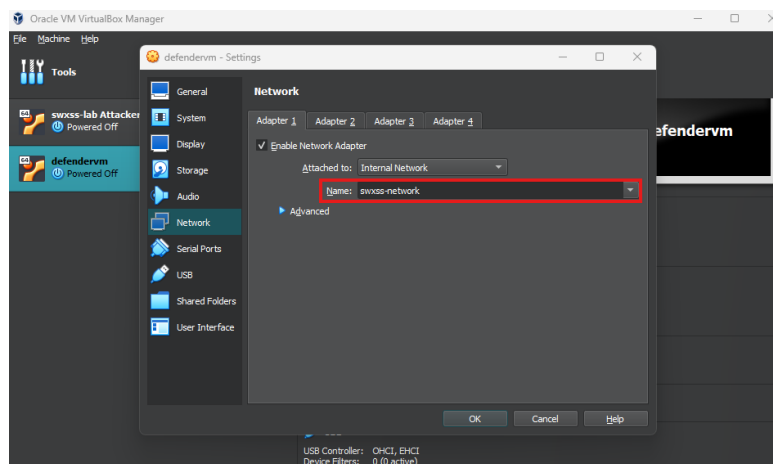


Figure 15: Change name of the internal network.